



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY

ICT 1209 -WEB TECHNOLOGIES
FINAL PROJECT REPORT

STUDENT RESOURCE HUB

Theme: Educational Web Application

Submitted by,

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1. PROJECT OVERVIEW

1.1 Project Overview

The Student Resource Hub is a database-driven web application developed as an educational web application for university students. The main purpose of the system is to provide a simple platform where students can access and manage their academic study materials in one place. Instead of searching for notes and learning resources from different sources, users can access the required resources through the Student Resource Hub.

The system allows users to create an account, log in to the system, search for study materials by subject or keyword, browse available resources, and send messages to the administrator through the contact form. The application is designed with a simple and user-friendly interface to make academic resources easier to access.

1.2 Purpose

The purpose of the Student Resource Hub is to develop a web-based system that helps students organize and access academic resources more easily. The system provides a centralized platform for accessing study materials and communicating with the administrator.

1.3 Target Audience

The main target users of the Student Resource Hub are:

- **University students**
- **Lecturers**
- **Academic staff**

1.4 Key Features

The main features of the Student Resource Hub are:

- **User Registration and Login** – Users can create an account, log in using their credentials, and log out from the system.
- **Resource Management** – Students can view and browse available study materials according to different subjects.
- **Search Function** – Users can search for notes and learning resources using a subject name or keyword.
- **Contact Form** – Students can send inquiries, suggestions, or feedback to the administrator.
- **Responsive Design** – The website is designed to be usable on desktop computers, tablets, and mobile phones.

2. SYSTEM DESIGN

2.1 Database Schema

The Student Resource Hub uses PHP and MySQL for database-related processing. The supplied backend code confirms a users table used for login and a messages table used for contact-form submissions.

The login backend searches the users table using the email field and verifies the stored password. The contact backend inserts the submitted name, email and message into the messages table.

Student Resource Hub – Database Schema

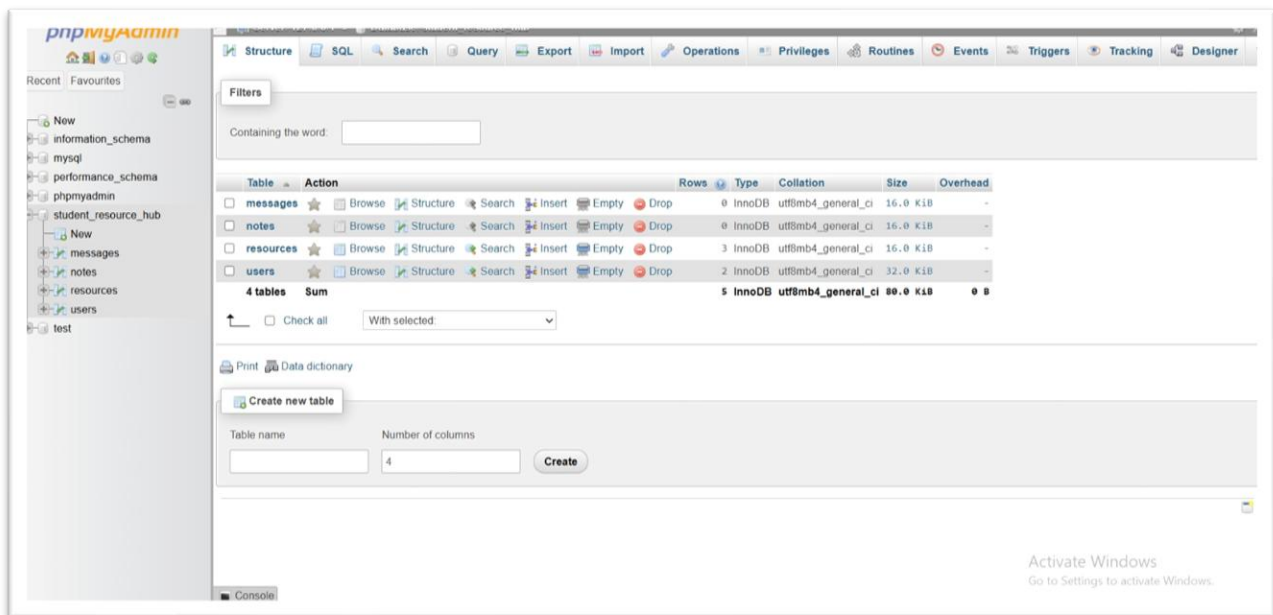
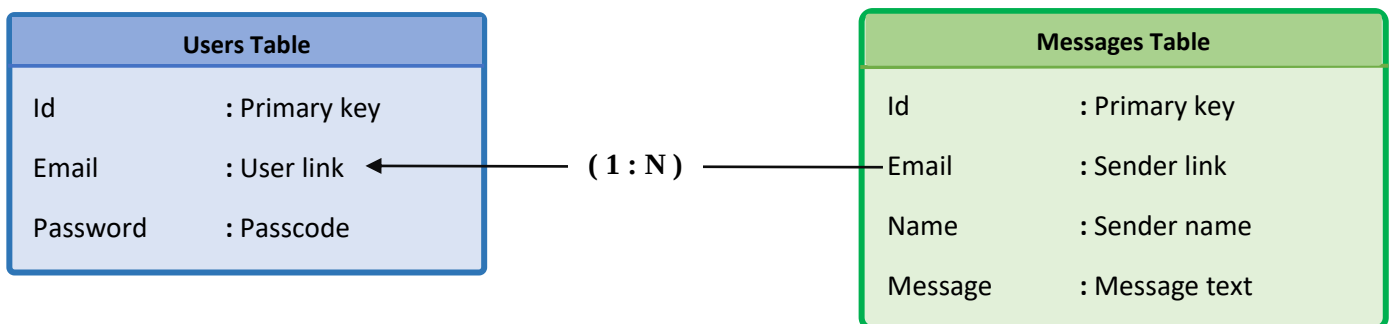


Figure 2.1: Database Entity-Relationship Diagram



2.2 Folder Structure

The project is organized into HTML pages, CSS files, an image folder and a backend folder. The backend folder contains the PHP files used for database connection, login processing and contact-message processing.

Student Resource Hub – Folder Structure

student resource hub/

- |— index.html
- |— contact.html
- |— login.html
- |— style.css
- |— contact.css
- |— login.css
- |— image/
- |— backend/
 - | |— db.php
 - | |— login.php
 - | |— contact.php
- |— .vscode/

3. IMPLEMENTATION

Home Page

The home page is implemented in index.html. It contains navigation links, welcome content and an Explore Resources action. It also provides subject categories including ICT, Mathematics and English. A resource listing area is included for displaying learning resources.

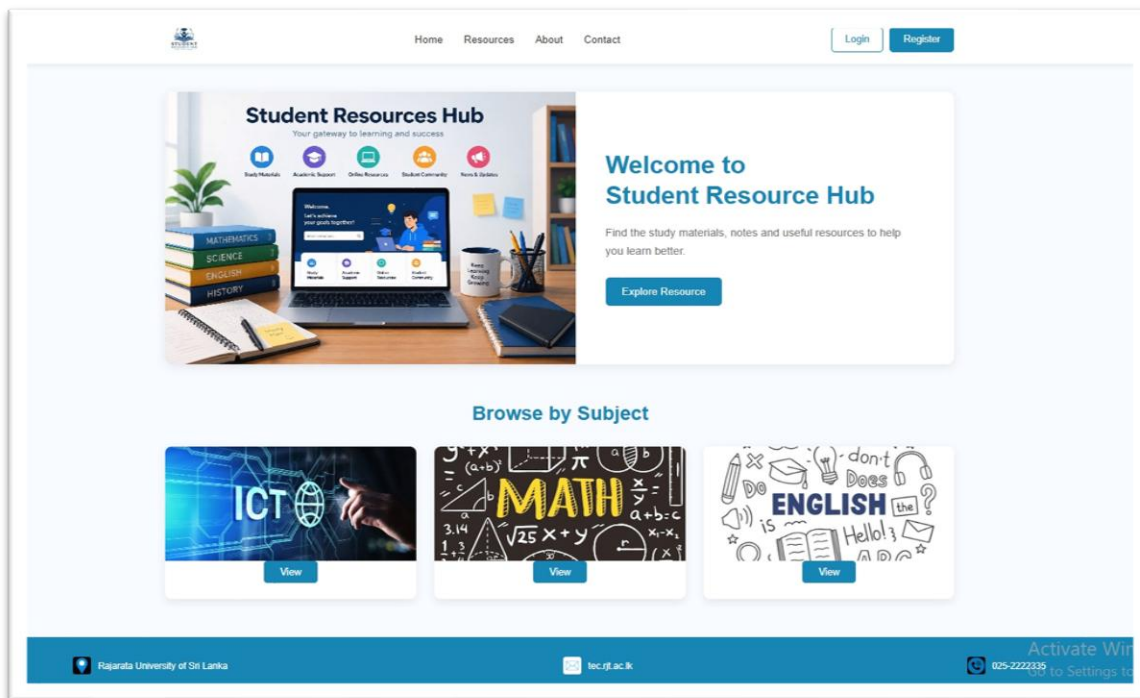


Figure 3.1: Student Resource Hub Home Page Interface

User Login

The login page is implemented in login.html. It provides email and password fields and sends the submitted data to backend/login.php using POST. The PHP backend uses a prepared SQL statement to find the user by email and uses password_verify to check the submitted password.

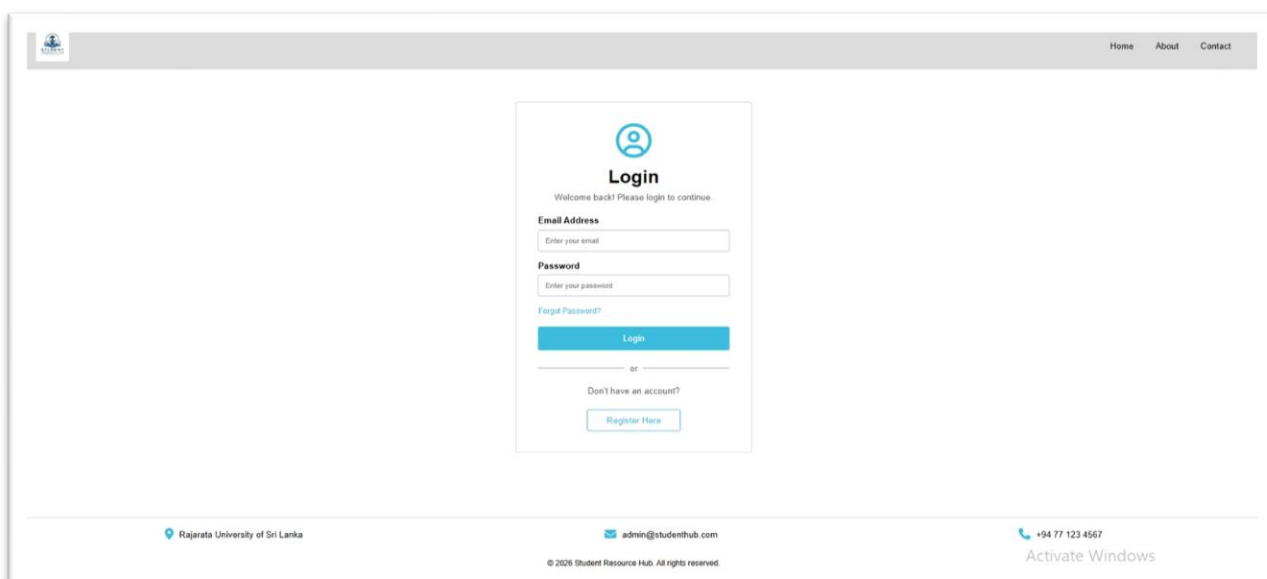
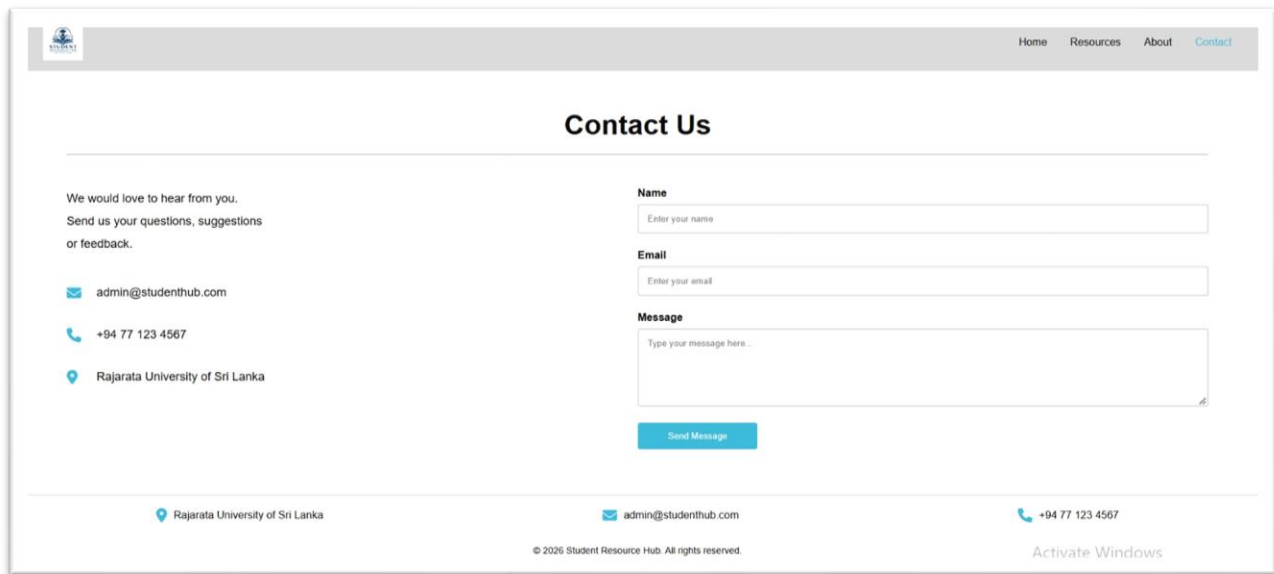


Figure 3.2: User Registration and Login Interface

Contact Form

The contact page provides fields for name, email and message. The form sends the submitted data to backend/contact.php. The backend uses a prepared statement to insert the values into the messages table.

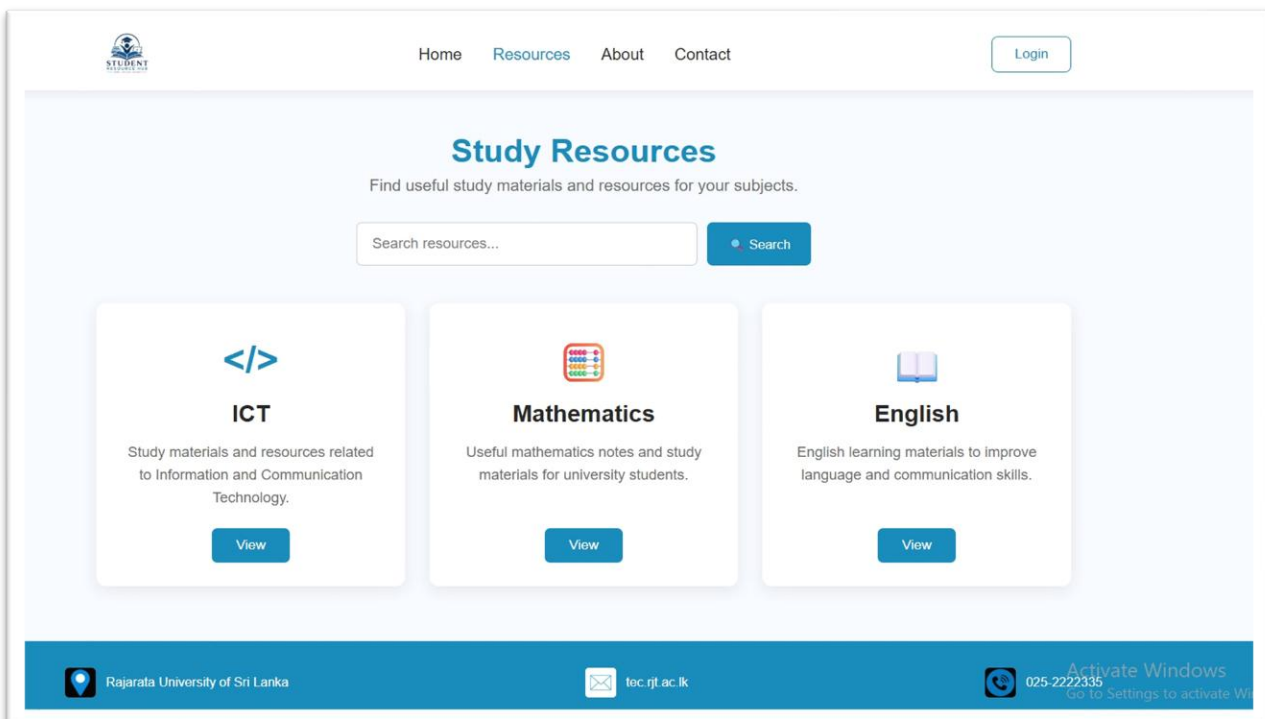


The screenshot shows a web page titled "Contact Us". On the left, there is a message: "We would love to hear from you. Send us your questions, suggestions or feedback." Below this, contact information is listed: an email icon followed by "admin@studenthub.com", a phone icon followed by "+94 77 123 4567", and a location pin icon followed by "Rajarata University of Sri Lanka". On the right, there are three input fields: "Name" with a placeholder "Enter your name", "Email" with a placeholder "Enter your email", and "Message" with a placeholder "Type your message here...". Below these fields is a blue "Send Message" button. At the bottom of the page, there is a footer with the same contact information as the left side, a copyright notice "© 2026 Student Resource Hub. All rights reserved.", and an "Activate Windows" watermark.

Figure 3.3: Administrator Contact Form Interface

Subject and Resource Browsing

The home page presents ICT, Mathematics and English as subject categories. The resource section is designed to display resource information.



The screenshot shows a web page titled "Study Resources". At the top, there is a navigation bar with links for "Home", "Resources", "About", and "Contact", and a "Login" button. Below the navigation bar, there is a search bar with the placeholder text "Search resources..." and a blue "Search" button. The main content area features three subject categories, each in a white box with a blue border. The first box is for "ICT" with a code icon, the second for "Mathematics" with a math icon, and the third for "English" with a book icon. Each box contains a brief description of the resources and a blue "View" button. At the bottom, there is a blue footer bar with the Rajarata University of Sri Lanka logo, the email "lec.rjt.ac.lk", the phone number "025-2222335", and an "Activate Windows" watermark.

Figure 3.4: Subject Categories and Resource Browsing Interface

Responsive Design

The CSS files include media queries for smaller screen sizes. The home page adjusts its navigation, welcome section, subject layout, resource rows and footer for smaller screens. The login page also contains responsive rules.

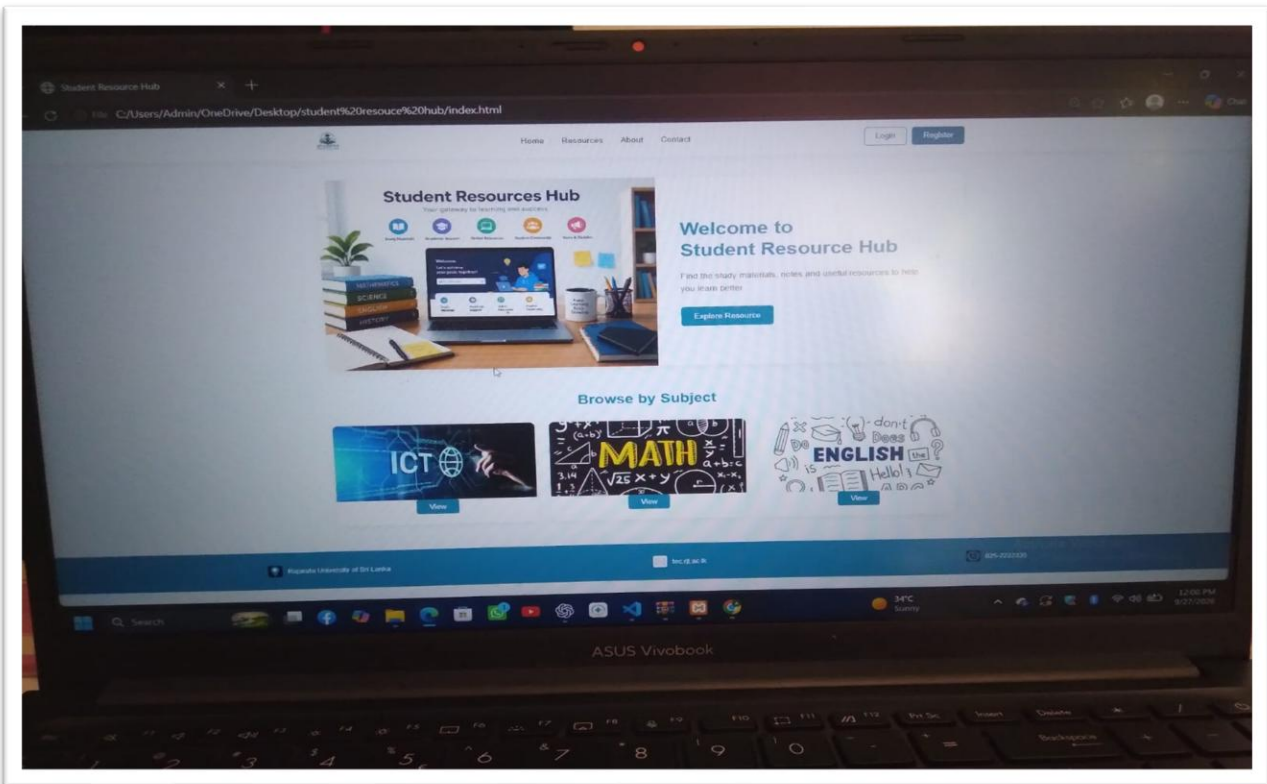


Figure 3.5: Mobile Responsive View of the Web Application

4. CHALLENGES AND SOLUTIONS

Challenge	Solution / Implementation
Connecting HTML forms to PHP backend	Forms were configured with POST actions pointing to PHP backend files such as login.php and contact.php.
Database interaction	Prepared SQL statements are used when processing login and contact form data.
Password verification	The login backend uses password_verify to compare the submitted password with the stored password value.
Responsive layout	CSS media queries were added to adapt the page layout for smaller screen sizes.
Resource loading	JavaScript fetch() API was used to load resource data dynamically from the PHP backend.

5. INDIVIDUAL CONTRIBUTION

The project was developed as a two-member group.

Group Member	Individual Contribution
P. R. N. Lakshani – 2745	• Developed and implemented the frontend of the Student Resource Hub. • Prepared and organized the final project report, including the required project documentation. • Collaboratively designed and refined the user interface using Figma with the other team member. • Actively participated in system testing, team discussions, problem solving, and overall project development.
A. K. M. B. Bandara – 2700	• Prepared the project proposal and planned the main project requirements. • Developed the backend using PHP and integrated it with the MySQL database. • Implemented the main backend functions such as user registration, login, and contact form processing. • Worked together on Figma design, testing, debugging, and overall project improvements.

6. SETUP INSTRUCTIONS

The project is intended to run in a local XAMPP environment using Apache and MySQL.

- Install and open XAMPP.
- Start Apache and MySQL from the XAMPP Control Panel.
- Copy the Student Resource Hub project folder into the XAMPP htdocs folder.
- Open phpMyAdmin and create or import the project database.
- Check that the database connection settings in backend/db.php match the local MySQL configuration.
- Open the project through localhost in a web browser.
- Test the home page, login form and contact form.

Database Import



student_resource_hub
.sql

7. REFERENCES

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